
Physik III

Autumn Semester 2026

Lecture Notes

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References

[1] Home, Jonathan: Vorlesungsskript Physik III. ETH Zürich. 2026

Physik III

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Lec 1

1 Optics

Optics is the study of light. It is relevant as light is everywhere. Furthermore it is a key tool for measurements in science. Furthermore, it introduces ideas associated with quantum mechanics. Historically, there was a wave and a corpuscle¹ picture of light. This wave/particle duality is the essence of quantum mechanics.

1.1 Fermats Principle

Fermat's principle tries to describe the path of light.

Proposition 1.1: Fermat's Principle

There are two different statements of Fermat's principle.

"Light travels by the route which takes the shortest time."

"Light will take the path which is extremal with respect to variations in the optical path"

The first statement is more intuitive, but the second statement is more general. For example, consider a setup with a mirror

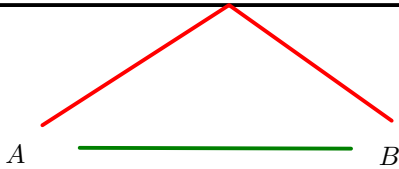


Figure 1: Two people seeing each other with a mirror

Definition 1.2: Optical Path

The **OPTICAL PATH** is a measure of time taken, written as distance.

$$D = ct = \frac{cs}{v}.$$

Where c is the speed of light and t is the time taken.

Recall that the speed of an electromagnetic wave in a medium is given by $v = \frac{c}{n}$, where n is the refractive index of the medium. Thus we can write the optical path as

$$D = \frac{cs}{v} = ns.$$

¹Light comes as particles

If the light has a certain frequency f , it will oscillate ever $t_c = \frac{1}{f}$. So the number of cycles in a time t is given by

$$N = \frac{t}{t_c} = \frac{fD}{c} = \frac{D}{\lambda_0},$$

where λ_0 is the **FREE-SPACE WAVELENGTH** of light.

The optical path tells us how many cycles of oscillation we have covered on our path.

We now want to discover the refraction law using Fermat's principle.

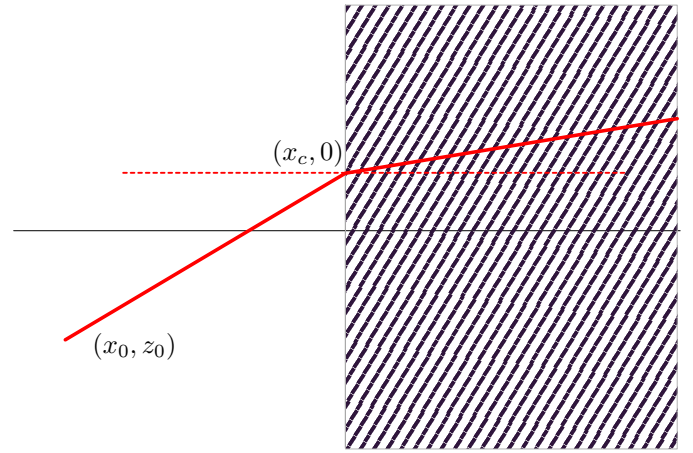


Figure 2: Derivation of Snell's law using Fermat's principle

The optical path in the left half is given by

$$D_1 = n_1 \sqrt{(x_c - x_0)^2 + z_0^2}.$$

In the right half, we have

$$D_2 = n_2 \sqrt{(x_p - x_c)^2 + z_p^2}.$$

So we want to minimize the total optical path $D = D_1 + D_2$ with respect to x_c . We have

$$\begin{aligned} \frac{dD}{dx_c} &= \frac{dD_1}{dx_c} + \frac{dD_2}{dx_c} = 0 \\ \frac{n_1}{\sqrt{(x_c - x_0)^2 + z_0^2}}(x_c - x_0) &= \frac{n_2}{\sqrt{(x_p - x_c)^2 + z_p^2}}(x_p - x_c) \\ n_1 \sin(\theta_1) &= n_2 \sin(\theta_2). \end{aligned}$$

Where θ_1 and θ_2 are the angles of incidence and refraction, respectively. This is known as Snell's law.

Theorem 1.3: Snell's Law

A light ray incident on a boundary between two media with refractive indices n_1 and n_2 will be refracted according to the law

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2).$$

1.2 Lenses

The best way to think of a lens is to think about the eye. It consists in principle of a lens and a screen (the retina). In principle, the lens is of refractive index n_2 in some sort of substance of refractive index n_1 . We are restricted by the shape and the refractive indices of the lens. Furthermore, we are restricted by the size of the lens.

Let us focus on the ideal lens.

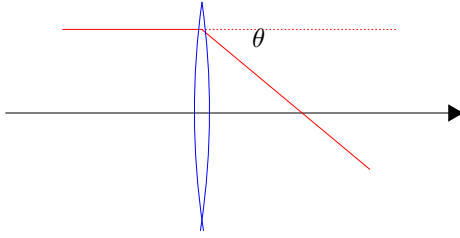


Figure 3: Ray bending of an ideal lens.

Light hitting the lens at a height h is deflected by the angle $\theta = \frac{h}{f}$, where f is the **FOCAL LENGTH** of the lens. Light incoming parallel to the optical axis will cross the optical axis at a distance f after the lens. The intersection is called **FOCAL POINT**. Here we make use of the **PARAXIAL APPROXIMATION**, which states that we only consider rays close to the optical axis, so we can use

$$\sin(\theta) \approx \tan(\theta) \approx \theta.$$

We will see, that in the paraxial approximation, if we have a non parallel ray incoming, it will still focus a distance f after the lens. In general, parallel rays converge after the lens in the **FOCAL PLANE**.

We now want to trace rays through a single lens.

By similar triangles, we have

$$\frac{\Delta}{\delta} = \frac{d}{s} =: M.$$

where M is the **MAGNIFICATION** of the lens. By similar triangles again, we have

$$\frac{\delta}{f} = \frac{\Delta}{d-f} \Rightarrow \frac{d-f}{f} = \frac{\Delta}{\delta} = \frac{d}{s}.$$

Hence,

$$\frac{d}{f} - 1 = \frac{d}{s} \Rightarrow \frac{1}{f} = \frac{1}{s} + \frac{1}{d}.$$

This is called the **LENS EQUATION**.

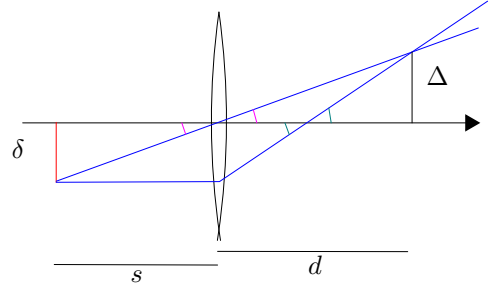


Figure 4: Derivation of the lens equation using ray tracing.

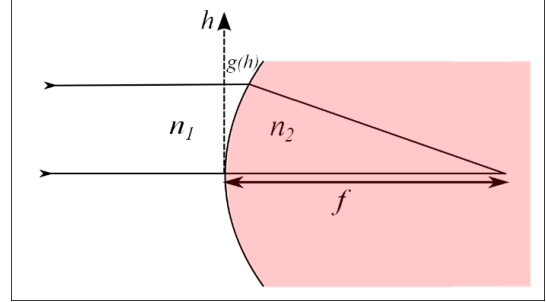


Figure 5: Setup of the surface lens.

Let us figure out how to design a simple "surface" lens. For this, we describe the profile by $g(h)$.

The optical path is given by

$$D = n_1 g(h) + n_2 \sqrt{h^2 + (f - g(h))^2}.$$

In general this is not really solvable, so we consider the paraxial approximation. So we have $h \ll f$. Since the lens should be symmetric we can assume

$$g(h) = \sum_n a_{2n} h^{2n}.$$

The simplest solution would be $g(h) = a_2 h^2$. Putting this into the expression, we get

$$\begin{aligned} D &= n_1 a_2 h^2 + n_2 \sqrt{h^2 + (f - a_2 h^2)^2} + O(h^4) \\ &= n_1 a_2 h^2 + n_2 \sqrt{h^2 + f^2 + a_2^2 h^4 - 2f a_2 h^2} + O(h^4). \end{aligned}$$

Using that $f \gg a_2 h^2, h$ we get

$$D = n_1 a_2 h^2 + n_2 f \sqrt{\frac{h^2}{f^2} + 1 + \frac{a_2^2 h^4}{f^2} - \frac{2a_2 h^2}{f}} + O(h^4).$$

neglect

Using the Taylor expansion $\sqrt{1+x} = 1 + \frac{x}{2} + O(x^2)$, we get

$$D = n_1 a_2 h^2 + n_2 f \left(1 + \frac{h^2}{2f^2} - \frac{a_2 h^2}{f} \right) + O(h^4).$$

Reordering the terms, we get

$$D = n_2 f + \left(n_1 a_2 + \frac{n_2}{2f} - n_2 a_2 \right) h^2 + O(h^4).$$

We want all paths to have the same optical path. So we want the coefficient of h^2 to vanish. This gives us

$$n_1 a_2 + \frac{n_2}{2f} - n_2 a_2 = 0 \Rightarrow a_2 = \frac{n_2}{2f(n_2 - n_1)}.$$

Unfortunately it is very annoying to make a lens with a parabolic profile. A spherical profile would be much easier.

Lenses in general are imperfect. Possible sources are:

1. Ignored higher order terms: symmetric about axis $\Rightarrow$ spherical aberration
2. Off axis aberrations: coma.
3. Chromatic aberrations: The refractive index depends on the wavelength of light. So different colors focus at different points.

Lec 2

Let us look at some imaging instruments in more detail. The human eye is limited by: the size of the eye is fixed. As we have seen

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{d}.$$

For the human eye, d is fixed. Secondly, the range of focal lengths is limited. Let us consider the telescope, using lenses to help. The main problem we want to solve is that we want to see two stars subtending a small angle θ .

If we have only one lens, since the angle is small, we have

$$\theta = \frac{\delta}{s} = \frac{\Delta}{d}.$$

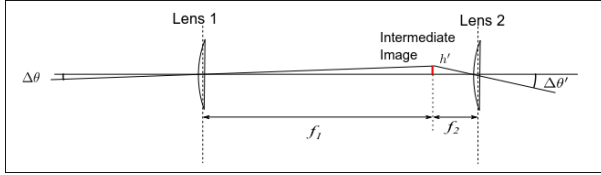


Figure 6: Telescope with two lenses.

Introducing a second lens, we first see, that image 1 is at f_1 behind lens 1, since

$$\frac{1}{f_1} = \frac{1}{s} + \frac{1}{d} \approx \frac{1}{d},$$

since s is very large. For the second lens, we then find

$$\theta_2 = \frac{\Delta}{f_2} = \frac{d\theta}{f_2} = \frac{f_1}{f_2} \theta.$$

Nowadays, we use a telescope with a mirror instead of a lens. The advantage is that we don't get chromatic aberration, since the mirror reflects all wavelengths the same way.

For looking at very small things, we typically use a microscope. Recall the magnification

$$M = \frac{\Delta}{\delta} = \frac{d}{s}.$$

So if we want to see very small things, we need to make s very small. The problem is that we need to satisfy the lens equation.

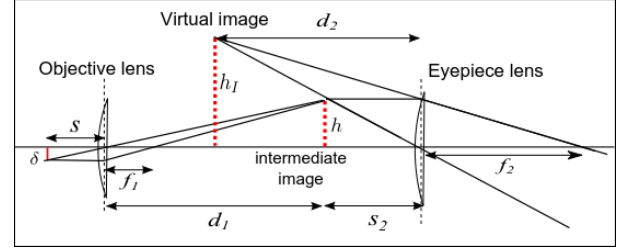


Figure 7: Microscope with two lenses.

Using a lens with $f_1 < s$ we thus get a magnification

$$h = \frac{d}{f_1} \delta.$$

In general, propagating through lenses is done, by finding the image through the first lens and then propagate that image through the second lens.

Typically in a microscope, we will use a lot of lenses. This is mainly because then we can only change the angle a little bit and hence reduce the spherical aberration. It also helps out cancelling out chromatic aberration.

1.3 Wave Optics

Starting from Maxwell's equations, we can turn them into a wave equation

$$-\nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}.$$

To simplify the problem, we can decide to treat directions of $\vec{E}$ separately. Then for example,

$$-\frac{\partial^2 E_x}{\partial x^2} = -\mu_0 \epsilon_0 \frac{\partial^2 E_x}{\partial t^2}.$$

This is the scalar wave equation.

Measuring the light typically involves measuring the intensity, given by

$$I = \frac{cn\epsilon_0 |\vec{E}|^2}{2},$$

which is measured as power per unit area.

Let us look at the solutions of the scalar wave equation. It is given by

$$\nabla^2 V - \frac{1}{c^2} \frac{\partial^2 V}{\partial t^2} = 0.$$

We will use spherical coordinates, where the general solution is given by

$$V(r, t) = \frac{a}{r} \cos(kr - \omega t) + \frac{b}{r} \cos(kr + \omega t).$$

Here the first part represents an outgoing wave and the second part represents an incoming wave. Using complex numbers, we can write this as

$$V(r, t) = \frac{a}{r} e^{i(kr - \omega t)} + \frac{b}{r} e^{i(kr + \omega t)}.$$

We select the first term for a light source at the origin. Keep in mind that we can use complex numbers, but we can only measure real quantities. So we should always

finish with $\text{Re}(V)$ or $|V|^2$. We now want to look at the propagation of light through an aperture.

For this we consider a point source at (x_s, y_s, z_s) . We then place a screen at $z = 0$ with a hole in it. We then want to find the field at a point of interest (x_p, y_p, z_p) .

Since we have a point source, the outgoing wave, emitted by the point source is given by

$$V_{\text{emit}} = A \frac{e^{i(ks - \omega t)}}{s},$$

where $s = \sqrt{(x - x_s)^2 + (y - y_s)^2 + (z - z_s)^2}$. To propagate this wave through the hole, we use **HUYGEN'S PRINCIPLE**.

Proposition 1.4: Huygen's principle

Given an incoming wave, we can write down the wavefronts. Letting each point on the wavefront be a point source of a new wave, we can then find the outgoing wave by summing up all the contributions of the new waves.

For example a planar wave front, for each point on the wave front, we can consider a point source. The outgoing wave is then given by the sum of all the contributions of the point sources. Summing up this results in again a planar wave front. As a recipe, first take the original wave

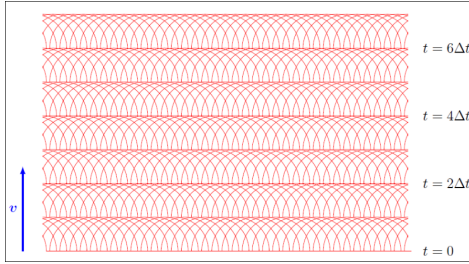


Figure 8: Huygen's principle for a planar wave front.

amplitude. The use this as a source of an outgoing wave at every position.

$$A \frac{e^{iks(x,y,z)}}{s(x,y,z)} \cdot \frac{e^{ikr}}{r},$$

where

$$r = \sqrt{(x_f - x)^2 + (y_f - y)^2 + (z_f - z)^2}.$$

Then, we sum up all the contributions of the outgoing waves.

For our problem, we constrain ourself to $x_f = x_p$, $y_f = y_p$ and $z_f = z_p$. Splitting the aperture into small pieces $dx dy$, we can write down the outgoing wave as

$$A \frac{e^{iks}}{s} dx dy \cdot \frac{e^{ikr}}{r}.$$

Summing up all the contributions, we get

$$U(P) \propto \iint \frac{e^{iks}}{s} \frac{e^{ikr}}{r} dx dy.$$

A more rigorous derivation would yield

$$U(P) = -\frac{iA}{4\pi} \oint \frac{e^{ik(s+r)}}{sr} (k\hat{r} \cdot \hat{n} - k\hat{s} \cdot \hat{n}) da.$$

Let us simplify the expression from Huygen's approach. First, since the aperture is small, compared to the distance to the source, we can approximate the amplitude as constant. I.e. if $s_0 \gg \lambda$, Δ_x, Δ_y , we can approximate $\frac{1}{s} \approx \frac{1}{s_0}$. So the incoming wave is given by

$$\frac{1}{s_0} e^{iks}.$$

Furthermore if the point of interest is far enough away, we can make the same approximation. Thus,

$$U(P) \propto \frac{A_0}{r_0 s_0} \iint e^{ik(s+r)} dx dy.$$

Defining the center of the aperture as $(0, 0, 0)$, we can write down the distances as

$$s_0 = \sqrt{x_s^2 + y_s^2 + z_s^2}, \quad r_0 = \sqrt{x_p^2 + y_p^2 + z_p^2}.$$

The interpretation of the integral is given as the amplitude is given by sum over amplitudes due to all paths. For this we simplify the expression. In general, the aperture size $\Delta_x, \Delta_y \ll |s_0|, |r_0|$. We have

$$\begin{aligned} r &= \sqrt{(x_p - x)^2 + (y_p - y)^2 + (z_p - z)^2} \\ &= \sqrt{x^2 + y^2 + z^2 + x_p^2 + y_p^2 + z_p^2 - 2(xx_p + yy_p + zz_p)} \\ &= \sqrt{x^2 + y^2 + r_0^2 - 2(xx_p + yy_p)} \\ &= r_0 \sqrt{1 + \frac{x^2 + y^2}{r_0^2} - 2\frac{xx_p + yy_p}{r_0^2}}. \end{aligned}$$

Using $x, y \ll r_0$, we get $\frac{x}{r_0} \ll 1 \Rightarrow \frac{x^2}{r_0^2} \ll 1$. Thus

$$r \approx r_0 \left(1 - \frac{xx_p + yy_p}{r_0^2} \right).$$

Analogous calculation can be done for s . Thus we get

$$s \approx s_0 \left(1 - \frac{xs_s + ys_s}{s_0^2} \right).$$

Hence,

$$U(P) = K \int_A e^{ik(s+r)} da = K e^{ik(s_0+r_0)} \int_A e^{-ik(xX+yY)} da,$$

where

$$X = \frac{x_s}{s_0} + \frac{x_p}{r_0}, \quad Y = \frac{y_s}{s_0} + \frac{y_p}{r_0}.$$

The approximation, that the phase is linear in x and y is called the **FRAUNHOFER APPROXIMATION**.

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